MODULE
# 6.2 Compounds and mixtures

The billions of substances in the universe are all made up of just 100 or so elements. By understanding how the elements combine to form new substances, scientists can develop new, useful materials such as plastics, perfumes and medicines. Scientists can also make natural materials such as vitamins, flavours and even blood in the laboratory.



## Compounds

Most substances you encounter every day are made up of more than one type of atom. These forms of matter are known as **compounds**. Every compound has a unique set of characteristic properties. The properties of compounds are usually very different from the properties of the elements that make them up. Some compounds found around your home are shown in Figure 6.2.1.

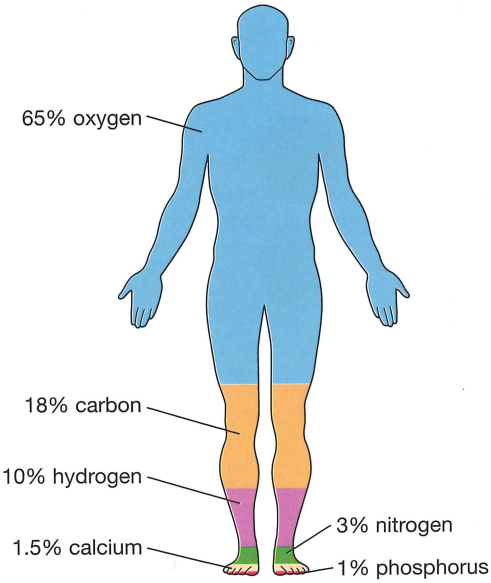



**FIGURE 6.2.1** Different compounds have different properties.

### SciFile

#### What are you made of?

The human body is made up of just 16 elements arranged into a huge variety of compounds. However, 98.5% of the body is made up of just six elements! These are oxygen (65%), carbon (18%), hydrogen (10%), nitrogen (3%), calcium (1.5%) and phosphorus (1%). This is shown in the human-shaped graphic below.



Some compounds are made up of molecules—such as water, wax and vegetable oil. In these compounds, each molecule contains two or more different types of elements. Other compounds form crystal lattices. For example, table salt is made of a lattice of sodium and chlorine atoms, while beach sand is a lattice of silicon and oxygen.

## Molecular compounds

All the molecules in a compound are identical in size and shape, and have the same number of atoms. Scientists represent the molecules of a compound using a chemical formula. The chemical formula for a molecular compound is also known as a molecular formula, because it describes how many of each atom make up the molecule. The number of each type of atom in the compound is written in subscript (written below the line).

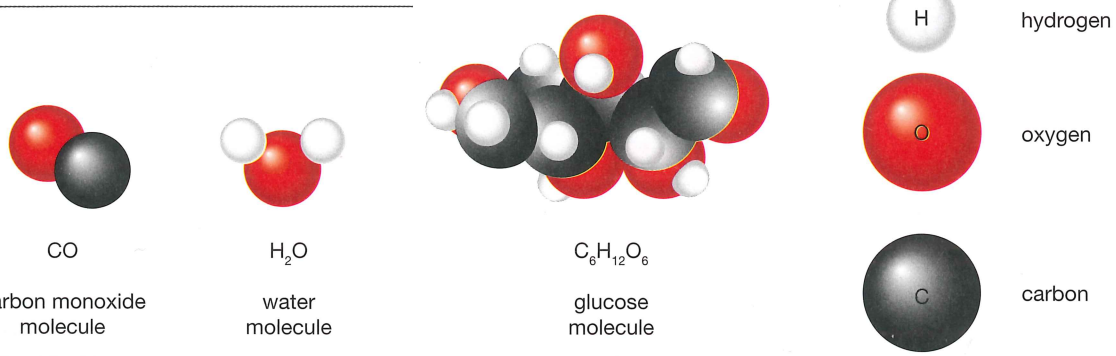
Figure 6.2.2 shows three examples of compounds made from different combinations of the same elements:

- Carbon monoxide has the molecular formula CO, which means it contains one carbon atom and one oxygen atom.
- Water is a molecular compound with the molecular formula $H_2O$. This molecular formula indicates that there are two hydrogen (H) atoms and one oxygen (O) atom in each molecule of water ($H_2O$).
- Glucose has the molecular formula $C_6H_{12}O_6$, which means it contains six carbon atoms, twelve hydrogen atoms and six oxygen atoms.

These three compounds are all are relatively small molecules, made up of just a few atoms. However, the molecules of other compounds, such as some plastics, contain thousands of atoms and can stretch several metres long.

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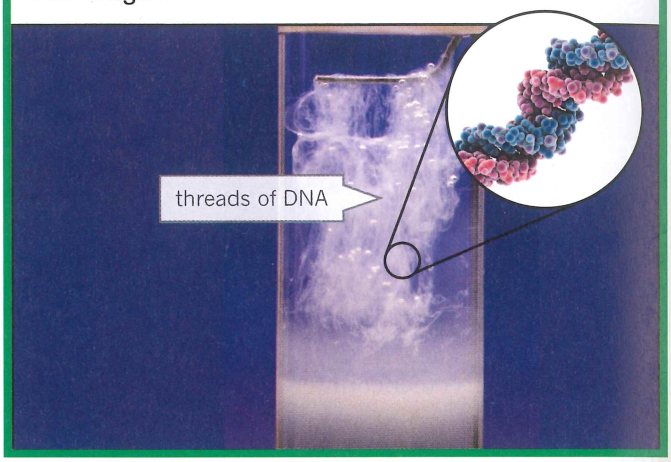
**FIGURE 6.2.2** Molecules of three common compounds. All of these compounds are made from just three elements—oxygen, hydrogen and carbon.




### SciFile

#### DNA

DNA (deoxyribonucleic acid) is a molecule found in the nucleus of a cell. DNA contains all the genetic information that makes you who you are. DNA is a very, very long molecule. The fine threads of DNA shown below would each be 2–3 metres long if stretched to their full length.




### SkillBuilder

#### Making sense of molecular formulas

The molecular formula of a compound indicates exactly what is in the compound's molecules.

The chemical symbols tell you what types of atoms are in the molecule.

$H_2O$

The subscript numbers tell you how many atoms of each type are in the molecule.

If there is no number, then it means there is only one atom.

# orked example

## aking sense of molecular formulas

### oblem 1

ntify the types and numbers of atoms in a lecule of nitric acid, $HNO_3$. The molecule is shown igure 6.2.3.

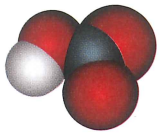


FIGURE 6.2.3

### lution

*nking: Identify the elements in the molecule by king at the chemical symbols.*

rking: The molecule contains hydrogen (H), rogen (N) and oxygen (O).

*nking: Identify the number of atoms for each ement by looking at the subscripts.*

rking: H and N have no subscripts, so there is one atom and one N atom in each molecule.

nas 3 as its subscript, so there are three O atoms each molecule.

### roblem 2

entify the types and numbers of atoms in a olecule of ethanol, $C_2H_5OH$. The molecule is shown Figure 6.2.4.



FIGURE 6.2.4

### olution

*hinking: Identify the elements in the molecule by oking at the chemical symbols.*

Vorking: The molecule contains carbon (C), ydrogen (H) and oxygen (O).

*hinking: Identify the number of atoms for each lement by looking at the subscripts.*

Vorking: C has the subscript 2, so there are two atoms.

has no subscript so there is one O atom.

appears in two places, so add the subscripts ogether: 5 + 1 = 6. So there are six H atoms in each nolecule.

### Problem 3

Construct a molecular formula for the sulfuric acid molecule shown in Figure 6.2.5.

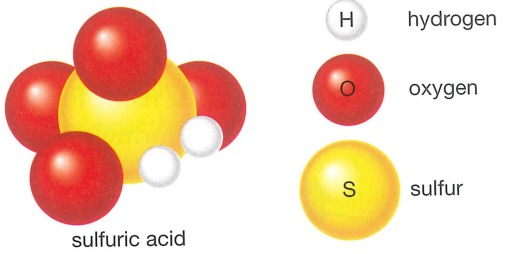



FIGURE 6.2.5

### Solution

*Thinking: Identify the elements in the molecule by looking at the key.*

Working: The molecule contains hydrogen (H), sulfur (S) and oxygen (O) atoms.

*Thinking: Identify the number of atoms for each element by counting how many of each type there are.*

Working: There are 2 white balls (H), 1 yellow ball (S) and 4 red balls (O) so the molecular formula for sulfuric acid is $H_2SO_4$.

### Try yourself

1 Identify the types and numbers of atoms in molecules of:
   - **a** benzene $C_6H_6$
   - **b** acetone $CH_3COCH_3$

2 Construct a molecular formula for the methanol molecule shown in Figure 6.2.6.

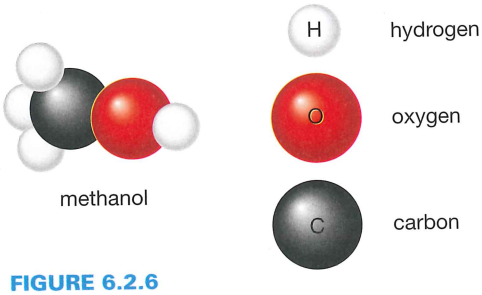



FIGURE 6.2.6

## Lattice compounds

Compounds can also form crystal lattices. In these lattices, the atoms are bonded very strongly to each other, so these compounds tend to be hard solids at room temperature. Two common examples of lattice compounds are table salt and beach sand.

The chemical formulas for lattice compounds are different from the molecular formulas used for molecules. For example, the lattice for table salt is shown in Figure 6.2.7. The scientific name for table salt is sodium chloride and its chemical formula is NaCl.

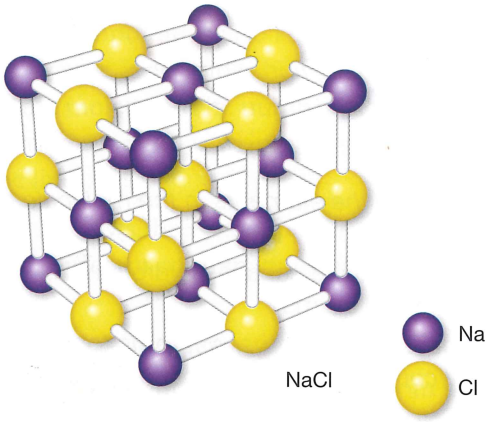



**FIGURE 6.2.7** In a lattice of sodium chloride, sodium and chlorine atoms form a grid-like structure with one sodium atom for every chlorine atom.

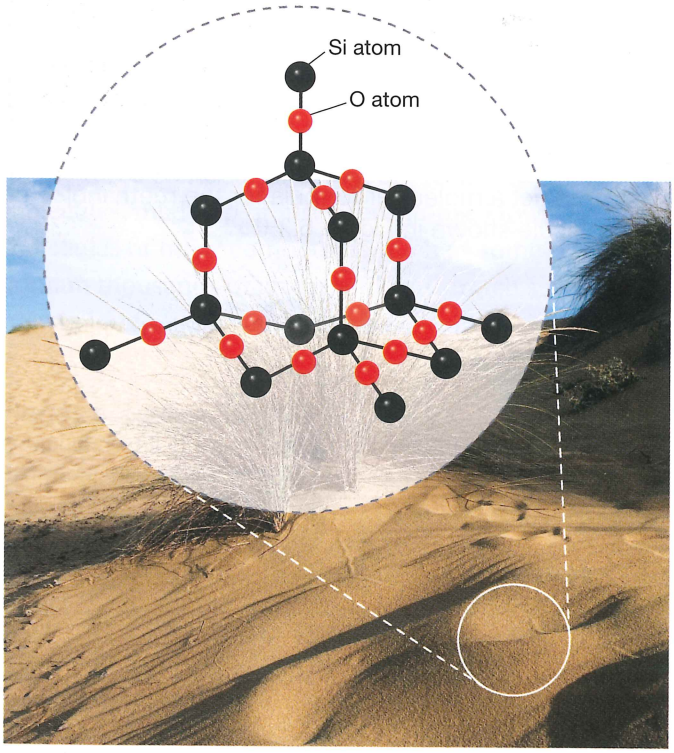



**FIGURE 6.2.8** Silicon dioxide is a compound lattice. Its formula indicates that there are two oxygen atoms for every silicon atom in the lattice.

From the chemical formula, you can see that table salt is made from sodium (Na) and chlorine (Cl). The formula also shows that there is one sodium for each chlorine atom in the crystal lattice.

Beach sand has the scientific name silica or silicon dioxide. Its chemical formula is $SiO_2$. This formula indicates that in a grain of beach sand, there is one silicon atom for every two oxygen atoms.

You can see the structure of silicon dioxide in Figure 6.2.8. Silicon dioxide also makes up minerals like quartz, amethyst and opal. It is the main ingredient in glass, and so it is used in optical fibres to carry information such as cable TV and broadband internet.

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## Mixtures

Many of the substances used every day are not simply elements or compounds but are **mixtures** of elements and compounds. A mixture is made up of more than one substance, where the substances are not chemically bonded together.

## STEM 4 fun

### Models from nuts and bolts




**PROBLEM**

Model elements, compounds and mixtures with nuts, bolts and washers.

**SUPPLIES**

- bolts, nuts and washers (that fit the bolts)

**PLAN AND DESIGN** Design the solution. What information do you need to solve the problem? Draw a diagram. Make a list of materials you will need and steps you will take.

**CREATE** Follow your plan. Construct your solution to the problem.

**IMPROVE** What works? What doesn't? How do you know it solves the problem? What could work better? Modify your design to make it better. Test it out.

**REFLECTION**

1. What area of STEM did you work in today?
2. In what career do these activities connect?
3. What did you do today that worked well? What didn't work well?

This means the substances that make up the mixture can be easily separated. For example, salt water is a mixture of table salt (sodium chloride) and water. You can separate the salt from the water by heating the saltwater, which causes the water to evaporate, leaving behind salt crystals. Air is a mixture of the elements oxygen ($O_2$) and nitrogen ($N_2$) with compounds such as water vapour ($H_2O$), carbon dioxide ($CO_2$) and carbon monoxide (CO). It is billions of molecules of these elements and compounds that you feel hitting your skin when you feel a breeze. Likewise, water from the tap is not pure but a mixture of water molecules ($H_2O$) and elements including chlorine, fluorine and many other elements in extremely small amounts (trace elements). Anything that has a list of ingredients, such as shampoo or soft drink, is a mixture.

## Everyday mixtures

There are many different types of mixtures that you use every day. One common type of mixture is a **solution**. A solution is formed when a solid dissolves in a liquid. In a solution:

- the dissolving solid is known as the **solute**
- the liquid in which the solute dissolves is known as the **solvent**.

Many common solutions have water as their solvent. These water-based mixtures are referred to as **aqueous solutions**. For example, sugar dissolves in water to form an aqueous sugar solution—sugar is the solute and water is the solvent.

However, there are everyday mixtures other than solutions. For example, smoke is a mixture of fine solid particles mixed with air.

## Working with Science

### COSMETIC SCIENTIST

The ingredients and mixtures that make up our personal care products, such as shampoo, sunscreen and lipstick, are carefully selected and tested by scientists who are trained in cosmetic chemistry. Cosmetic scientists are responsible for developing products that are safe to use on skin and hair and have the right consistency, scent and colour. There is a growing demand for organic products that have not been tested on animals. This demand fosters innovation and research into these products.

A cosmetic scientist is mostly based in a laboratory, where they use different combinations of ingredients to create new products, update existing products by altering the quantities of existing ingredients, and undertake testing and quality control (Figure 6.2.9). One of the challenges of a cosmetic scientist's job is formulating products that don't 'go off'. Some products, like perfume and sunscreen, will be used over months or even years, so it is important that they keep their qualities for a long period of time.

Cosmetic science can be studied through a Certificate in Advanced Cosmetic Science, a Diploma of Personal Care Formulation or a Bachelor of Science (Chemistry). These qualifications will give you skills and knowledge in the chemistry and safety requirements of cosmetics, the biochemical interactions between skin and hair and chemical mixtures, and the formulation and testing of cosmetic products.



**FIGURE 6.2.9** A cosmetic scientist's job involves mixing ingredients to create personal care products, such as lipstick, moisturiser and nail polish. This cosmetic scientist is working in a perfumery in France.

With these skills, you will be qualified to work for cosmetic manufacturers, as a researcher, or as a freelance cosmetic scientist, making and selling your own formulations.

#### Review

1. Cosmetic scientists need to have a very good understanding of the chemical properties of each of the ingredients they work with. Why do you think this is important?
2. List five to ten products that you use that would have been formulated by a cosmetic scientist.

The shaving foam in Figure 6.2.10 is another mixture—this one is made up of tiny gas bubbles contained in a liquid. Milk is an example of an everyday mixture where solid particles are suspended in a liquid. Although the particles in milk are solid, they are very small and so will never settle to the bottom of the liquid.



**FIGURE 6.2.10** Shaving foam is a mixture of gas bubbles in a liquid.

AB 6.7

## Describing mixtures

In a mixture there are a variety of substances, and the amount of each substance can vary. Therefore you cannot write chemical formulas for mixtures in the same way as you do for elements and compounds. Instead, mixtures can be often only be described as a list of ingredients, like the one shown in Figure 6.2.11.

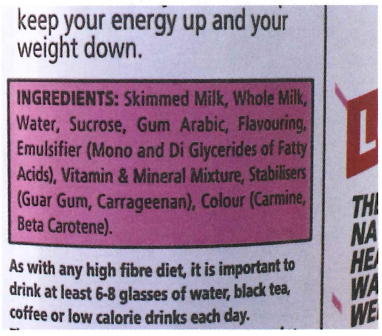



**FIGURE 6.2.11** Any substance made up of two or more ingredients is a mixture.

Solutions of substances dissolved in water are commonly used in chemistry and so these mixtures have been given more formal scientific names. For example, when copper(II) sulfate is dissolved in water, the solution is written as $CuSO_4(aq)$. In this formula, the (aq) stands for aqueous, which means dissolved in water. However, even this formula does not say exactly how much copper(II) sulfate is dissolved. For this reason, scientists must also say how concentrated a solution is. The concentration of a solution measures how much of a substance is in the solution. Concentration can be given in a number of ways. In the laboratory, concentration is usually expressed using a unit called molar (M). Hence, a 10 M solution of hydrochloric acid is much more concentrated (and dangerous) than a 0.1 M solution of hydrochloric acid.

## Alloys

**Alloys** are a mixture of a metal with other metals or non-metals. Alloys are particularly important because they often have very different and useful properties compared to the pure metal. For example, iron is cheap to produce but is soft and rusts easily, making it less useful. The properties of steel alloys vary greatly depending on what the iron is mixed with. This makes iron useful in a wide variety of applications. However, it is not possible to write a chemical formula for steel, because steel is a mixture of iron and other substances. Therefore, steel can only described by listing the substances in the alloy and how much there is of each substance, for example 3% carbon steel.

Adding small amounts of carbon to iron produces the alloy steel, which is strong and useful for building skyscrapers, bridges, machinery and cars. Adding chromium and nickel produces stainless steel, which stops it rusting. Some other common alloys are shown in Figure 6.2.12.




**FIGURE 6.2.12** Different mixtures of metals make different alloys with very different properties and a wide range of uses.

MODULE 6.2

# Review questions LS

## Remembering

1 Define the terms:
- a compound
- b mixture
- c solution.

2 What term best describes each of the following?
- a a solid that dissolves in a liquid to form a solution
- b a large network of atoms strongly bonded to each other
- c a mixture of two metals.

3 What are two examples of molecular compounds and two examples of lattice compounds?

4 a List three alloys.
- b State a use for each of the three alloys.

5 What are the chemical formulas for the following compounds?
- a carbon monoxide
- b glucose
- c water
- d silica (silicon dioxide).

6 List the elements and compounds that make up air.

## Understanding

7 Explain what the chemical formula tells you about:
- a a molecular compound
- b a crystal lattice compound.

8 Explain why a chemical formula cannot be written for a mixture.

9 Why is iron rarely used in its pure form?

10 a What is the chemical name for salt?
- b What is its chemical formula?
- c Is salt an example of a molecule or a crystal lattice?
- d Classify salt as an element or a compound.
- e Justify your classification.

11 The molecular formula for sulfuric acid is $H_2SO_4$. In this formula, what does the number 4 tell you?

12 Sodium oxide is a compound formed when sodium metal reacts with oxygen. It has the chemical formula $Na_2O$. In this formula, what does the number 2 tell you about the sodium oxide crystal lattice?

## Applying

13 What types of atoms are in the molecules below, and how many of each are there?
- a carbon dioxide, $CO_2$
- b nitrous oxide, $N_2O$
- c sucrose (sugar), $C_{12}H_{22}O_{11}$
- d trinitrotoluene (TNT), $C_7H_5N_3O_6$
- e acetic acid (vinegar), $CH_3COOH$
- f urea, $NH_2CONH_2$

## Analysing

14 Classify the following as element, compound or mixture.
- a pool water
- b helium
- c air
- d nitrogen
- e gold
- f sugar
- g cordial
- h carbon dioxide.

15 Classify the following compounds as either molecular or lattice.
- a water
- b carbon monoxide
- c silicon dioxide
- d carbon dioxide.

16 Determine the chemical formula for the molecule of methane shown in Figure 6.2.13.

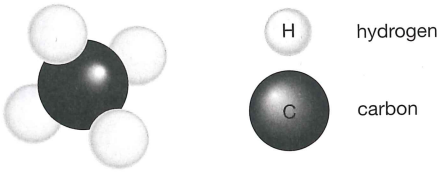
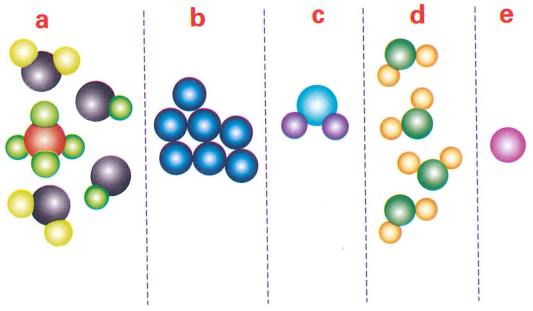



FIGURE 6.2.13

17 Match the following terms to the pictures below: *atom, molecule, element, compound, mixture.*




## Evaluating

18 Contrast:
- a elements and compounds
- b mixtures and compounds.